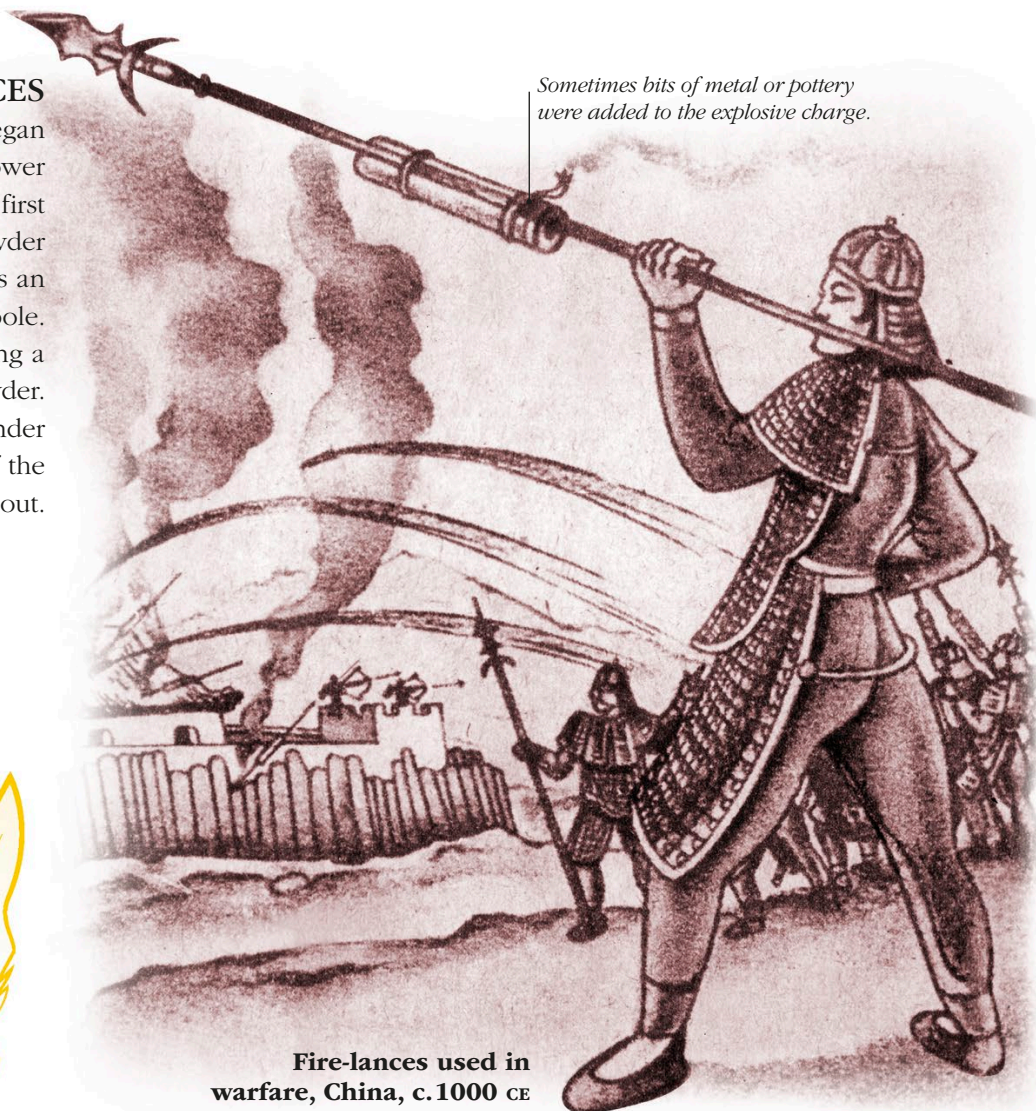


FIRE-LANCES

In around 950 CE, people began to understand the destructive power of gunpowder. One of the first weapons propelled by gunpowder was the fire-lance, which was an explosive charge attached to a pole. The charge was created by filling a bamboo cylinder with gunpowder. When lit with a fuse, the cylinder would explode in the direction of the enemy, sending flames shooting out.

WOW!

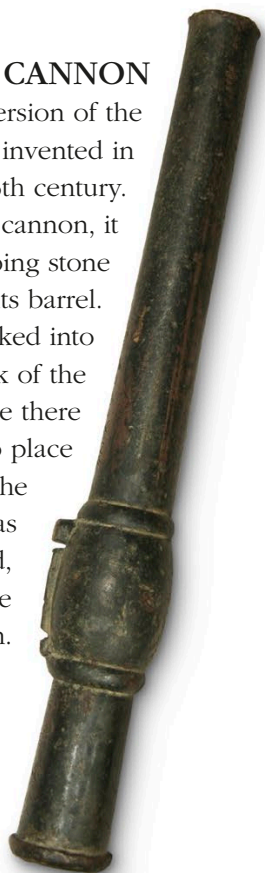
Adding metals to the gunpowder in fireworks makes sparks of different colors: copper for blue, barium for green, calcium for orange.



Fire-lances used in warfare, China, c.1000 CE

HAND CANNON

The earliest version of the handgun was invented in China in the 13th century. Known as a hand cannon, it was loaded by dropping stone or iron balls down its barrel. Gunpowder was packed into a chamber at the back of the hand cannon, where there was a small hole to place a lit fuse. When the gunpowder was ignited, it exploded, blasting out the ammunition.



Chinese bronze hand cannon, 1424

GREEK FIRE

The Chinese led the medieval world with explosive weapons, but they were not the first to use chemicals in warfare. The Byzantine Empire (centered in the area of modern-day Turkey and Greece) invented a substance called Greek Fire in around 672 CE. It was a thick, flammable liquid that burned even on water, making it deadly in sea battles. Its secret recipe probably included oil, sulfur, and saltpeter.



A 12th-century manuscript showing Byzantine sailors using Greek Fire